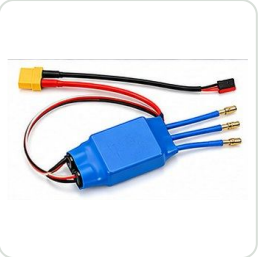
Electronic Speed Controller



Category: **Drones & Aviation** Type: **Short SOP** Priority: **Medium** 3 labs

Covers these lab items: Electronic Speed Controller; ESC (Electronic Speed controllers)

What it is used for

To install and check an electronic speed controller (ESC) in a compatible drone or motor training setup. To ensure correct wiring, configuration, and safe motor response before operation.

Main parts

- ESC electronic unit
- Battery power input leads
- Motor phase output leads
- Signal or control connector
- Cooling surface or heat sink, if fitted

Before you start

1. Confirm the ESC is compatible with the motor, battery voltage, and required current.
2. Inspect the ESC, wiring, connectors, and insulation for damage.
3. Disconnect the battery before making connections.
4. Review the ESC and motor manufacturer's wiring and configuration instructions.
5. Remove propellers before bench testing a drone motor.

How to use

1. Connect the motor leads to the ESC according to the manufacturer's instructions.
2. Connect the signal lead to the correct control output with the specified polarity.
3. Connect the battery input only after verifying polarity and connector compatibility.
4. Secure the ESC so it has appropriate cooling and cannot contact conductive parts.
5. Configure calibration and operating settings as specified by the manufacturer.
6. Perform a low-power test with the propeller removed and the area clear.
7. Disconnect the battery before changing wiring or correcting faults.

⚠ Safety

- Never connect a battery with reversed polarity.
- Keep fingers, clothing, and tools away from motor shafts.
- Do not operate an ESC with exposed damaged conductors or signs of overheating.
- Use a battery and power source within the ESC's rated limits.

Care & storage

- Keep the ESC dry and free of conductive debris.
- Inspect connectors, solder joints, and insulation regularly.
- Ensure cooling airflow is not blocked.
- Stop using the unit if it overheats, swells, or shows electrical damage.

Common mistakes to avoid

- Using an ESC with an incompatible battery or motor.
- Reversing power polarity.
- Testing with propellers attached in an unsecured setup.
- Ignoring overheating or loose electrical connections.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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